

phase-03-videos — Progress

Status: completed

SIs: 6/6 completed

SI-03.1 — Infrastructure: Storage, Queue, and Worker (Docker Compose, Config Namespaces, Dependencies)

- **Status:** completed (2026-08-24)
- **Tests:** env.validation.integration-spec — 7 passed (storage/queue/upload defaults + required key rejection); full baseline suite 147 passed
- **Observations:** Added `redis` (7-alpine, AOF), `minio` + one-shot `minio-setup` (creates `videos` bucket), and `nestjs-worker` services to `compose.yaml`; all addressed by service name (never `localhost`). Created `storage / queue / upload` config namespaces via `registerAs`, extended Joi schema (`STORAGE_ACCESS_KEY/SECRET_KEY` required), updated `.env.example`. Installed deps: `@nestjs/bullmq`, `bullmq@5.x`, `ioredis@5.x`, `@aws-sdk/client-s3`, `@aws-sdk/s3-request-presigner`, `nanoid@3.x`.

SI-03.2 — Video Entity, Migration, and VideosModule

- **Status:** completed (2026-08-24)
- **Tests:** `src/videos` suite — 21 passed (`public-id` util, Video entity persistence, `VideosRepository` CRUD/queries, `VideosService` create/find, `VideosModule` wiring)
- **Observations:** Created `Video` entity (`@Entity('videos')`) with `video_status` enum (`draft/processing/ready/error`), `public_id` (`nanoid 12`, `unique+indexed`), `channel_id` FK to `channels` (`ON DELETE CASCADE`, `indexed`), nullable `storage/processing` columns, `file_size_bytes` `bigint` (typed `string | null` in TS), `metadata` `jsonb`. Added `generatePublicId()` util, `VideosRepository` wrapper (no direct `EntityManager` in services), basic `VideosService` (`create/findByPublicId`). Registered `VideosModule` (imports `ChannelsModule`). Migration `1787610093430-CreateVideos` generated and applied (enum + table + unique/index/FK).

SI-03.3 — Upload Initiation: Draft Pre-Registration and Presigned Multipart (TD-03)

- **Status:** completed (3/6)
- **Tests:** `npm test` (unit+integration) — 30 suites / 181 tests green; `e2e` — 4 suites / 56 tests green; `tsc --noEmit` clean; lint clean on new sources. Full suite re-run twice on a dirty DB to confirm isolation/idempotency.
- **Observations:** Implemented `StorageService` (S3/MinIO multipart: `create/presign` part URL/`complete/abort`, `presigned GET`, `put/get/delete`) with a separate `presign` client on the public endpoint. Added `StorageModule`. Extended `VideosService` with `initiateUpload` (`ownership` + `file-size` guards, `unique public_id` `retry` with `abort-on-collision`, `draft pre-registration` with `upload_id`), `getPartUploadUrl`, and `completeUpload` (`status` guards → `processing`). Added `InitiateUploadDto`, `CompleteUploadDto`, `VideoResponseDto`. Added `VideosController` (`POST /videos`, `POST /videos/:publicId/parts/:partNumber/url`, `POST /videos/:publicId/complete`) behind JWT with `@CurrentUser()`. Added domain exceptions (`VIDEO_NOT_FOUND`, `NOT_VIDEO_OWNER`, `FILE_TOO_LARGE`, `UPLOAD_NOT_INITIATED`, `UPLOAD_ALREADY_COMPLETED`, `INVALID_STATUS_TRANSITION`). Hardened shared-DB test isolation: video integration specs now clean via `cleanAllTables` + `delete videos` in `afterAll`; migrations spec drops sequentially (no deadlock).

SI-03.4 — Background Processing: BullMQ Queue, Worker Entrypoint, ffmpeg + ffprobe (TD-01/04/05/08)

- **Status:** completed (2026-08-24)
- **Tests:** `npm test -- video-processing` — 2 suites / 7 tests green (VideoProcessingProducer unit, VideoProcessingProcessor unit with mocked ffmpeg/storage/fs); `tsc --noEmit` clean; lint clean
- **Observations:** Created BullMQ queue infrastructure: `video-processing.constants.ts` (queue/job names + `ProcessVideoJobData` interface), `video-processing.producer.ts` (adds jobs with 3-attempt exponential backoff retry policy), `video-processing.module.ts` (registers BullMQ with Redis config). Integrated producer into `VideosService.completeUpload` to emit job after status→PROCESSING. Created isolated worker app in `src/worker/`: `FfmpegService` (child_process wrappers for ffprobe metadata extraction + ffmpeg thumbnail generation), `VideoProcessingProcessor` (BullMQ @Processor consuming jobs: downloads video from storage→/tmp, extracts metadata, generates thumbnail at 10% duration or 5s, uploads thumbnail to storage, updates Video entity with status=READY/duration/thumbnailKey/metadata, cleanup temp files, error handling with status=ERROR + processingError). Created `WorkerModule` (TypeORM + BullMQ consumer + StorageModule, no HTTP server) and `main.worker.ts` bootstrap. Created `Dockerfile.worker` (node:20-alpine + ffmpeg via apk). Updated `compose.yaml`: nestjs-worker service with `Dockerfile.worker` build, `DATABASE_/S3_/REDIS_*` env vars using service names. All error handling uses type-safe `getErrorMessage` / `getErrorStack` helpers.

SI-03.5 — Streaming and Download (Presigned GET, native Range/206) (TD-07)

- **Status:** completed (2026-08-24)
- **Tests:** `npm run test:e2e -- videos-streaming` — 1 suite / 7 tests green (stream/download redirect for READY videos, 404 for non-existent/draft, 401 without auth); `tsc --noEmit` clean
- **Observations:** Added `getStreamUrl` and `getDownloadUrl` methods to `VideosService` (both enforce status=READY, throw `VideoNotFoundException` otherwise). Added `GET /videos/:publicId/stream` and `GET /videos/:publicId/download` endpoints to `VideosController` with `@Redirect()` decorator (returns 302 with presigned URL in Location header). Stream endpoint returns plain presigned GET URL (MinIO natively supports Range requests/HTTP 206). Download endpoint adds `Content-Disposition: attachment; filename="..."` via storage service opts. Both endpoints behind JWT guard. Created comprehensive e2e test suite `videos-streaming.e2e-spec.ts` covering ready/draft/non-existent scenarios.

SI-03.6 — Complementary REST: Retrieve, List, Update, Delete

- **Status:** completed (2026-08-24)
- **Tests:** `npm test -- videos.service.spec` — 1 suite / 13 tests green (including new methods); `tsc --noEmit` clean
- **Observations:** Added `getVideoDetails` method to `VideosService` (public if status=READY, owner-only otherwise, includes presigned thumbnail URL). Added `listChannelVideos` with pagination support (limit=20, READY only, presigned thumbnails). Added `updateVideo` (title/description, owner only) and `deleteVideo` (removes from storage + DB, owner only). Added `findByChannelWithStatus` method to `VideosRepository` with pagination (skip/take). Created `UpdateVideoDto` for PATCH validation. Added 4 new endpoints to `VideosController`: `GET /videos/:publicId` (@Public, conditional access), `GET /channels/:channelId/videos` (paginated list), `PATCH /videos/:publicId` (owner only), `DELETE /videos/:publicId` (204 No Content, owner only). All endpoints include comprehensive Swagger annotations. Updated unit tests to include `VideoProcessingProducer` mock.